ARTIFICIAL INTELLIGENCE

Lecture Set 18
Clustering

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Course Contents

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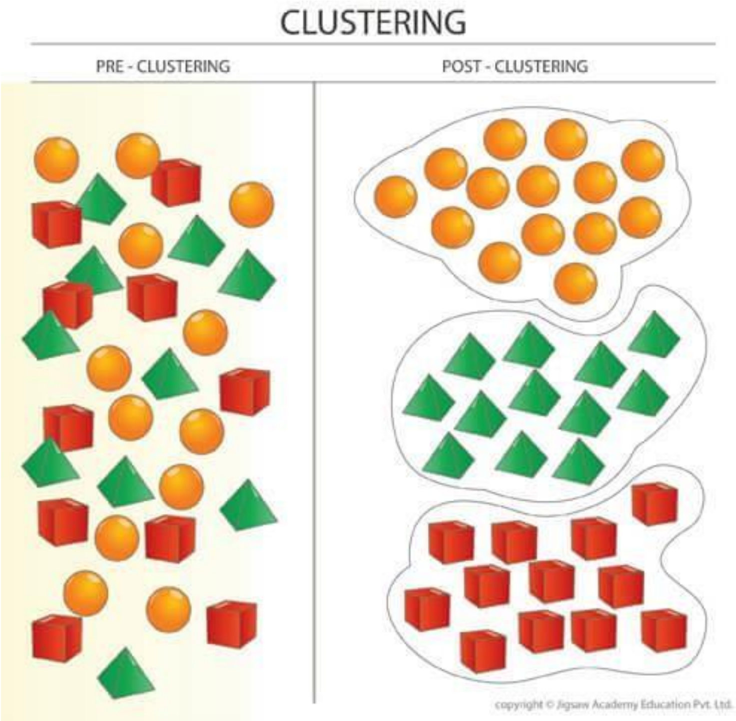
13. **Machine Learning (Supervised, Unsupervised, Reinforcement Learning)**

Clustering

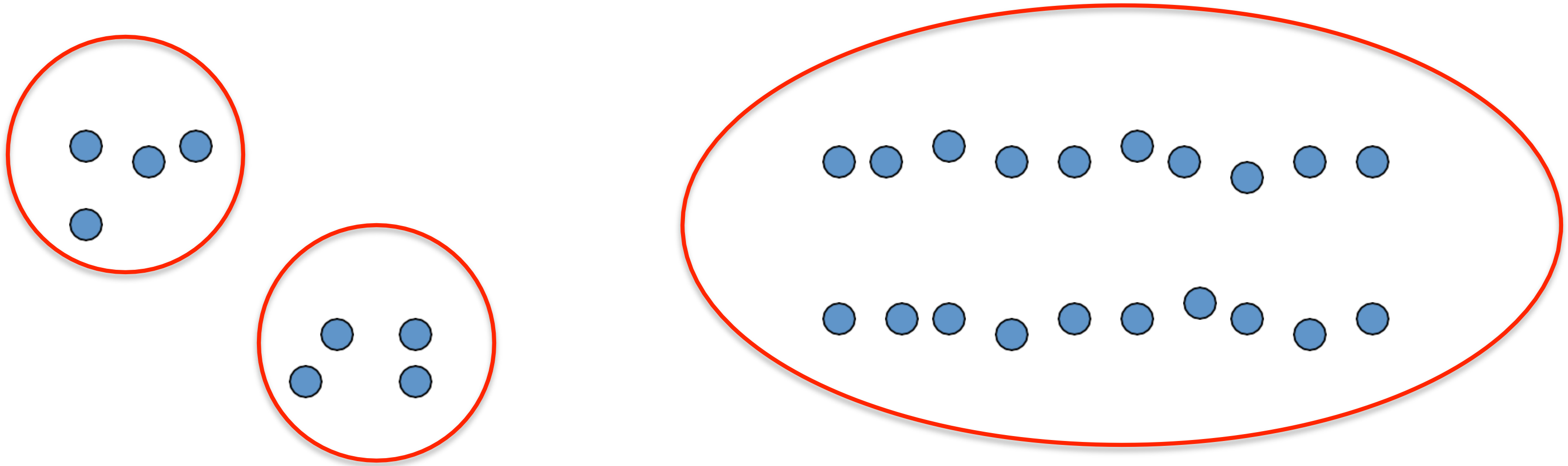
Clustering:

– Unsupervised learning
– Requires data, but no labels
– Detect patterns e.g. in
  • Group emails or search results
  • Customer shopping patterns
  • Regions of images
– Useful when don’t know what you’re looking for
– But: can get gibberish



Clustering

- Basic idea: group together similar instances
- Example: 2D point patterns



Clustering

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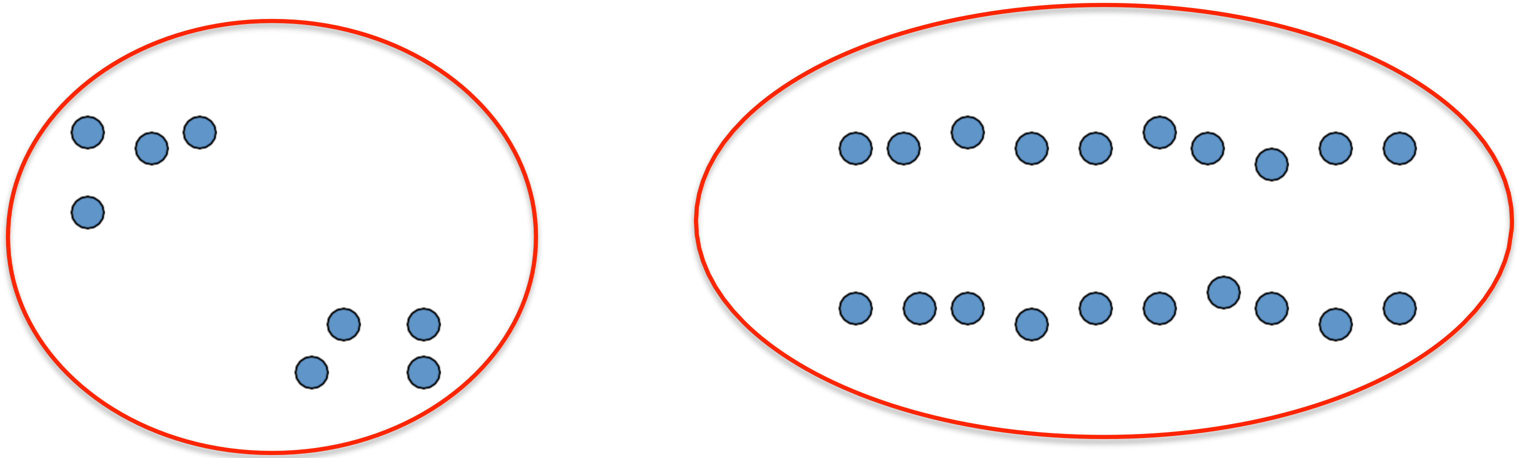
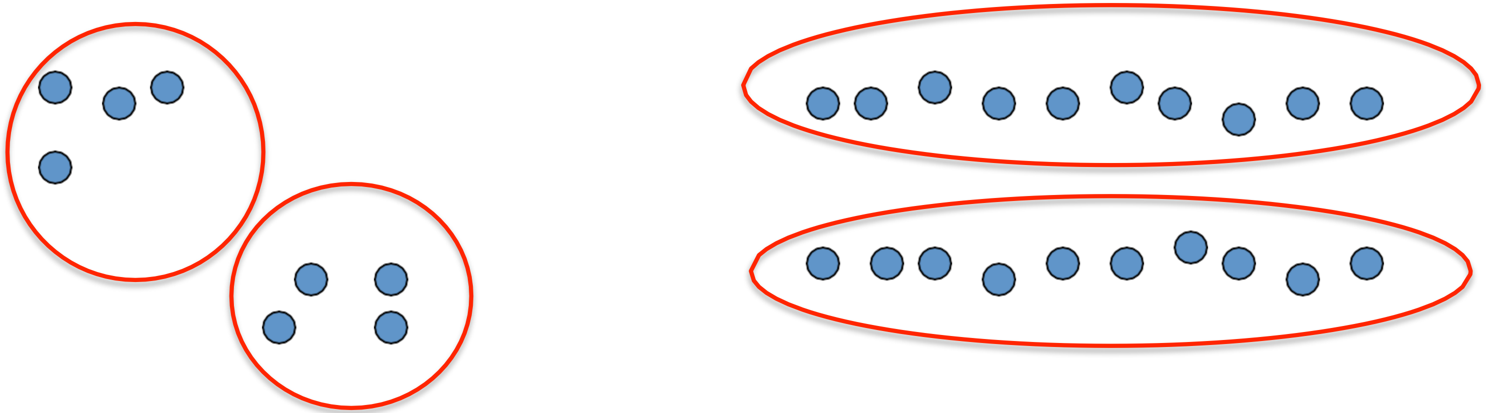
- Example: 2D point patterns



Clustering

- Basic idea: group together similar instances
- Example: 2D point patterns



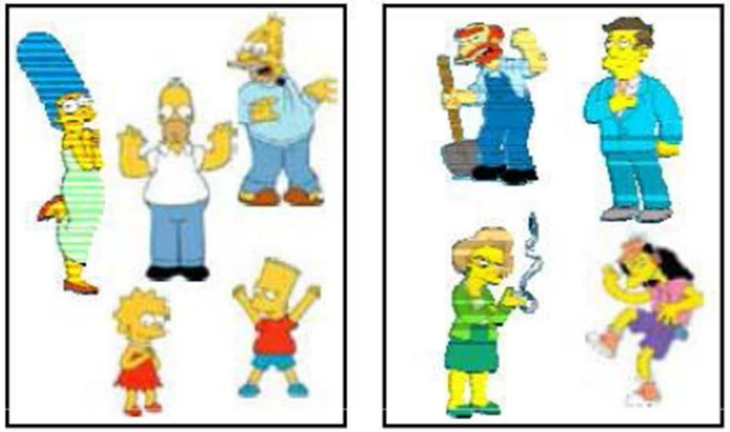
- What could "similar" mean?
  - One option: small Euclidean distance (squared)

$$dist(x, y) = ||x - y||_2^2$$

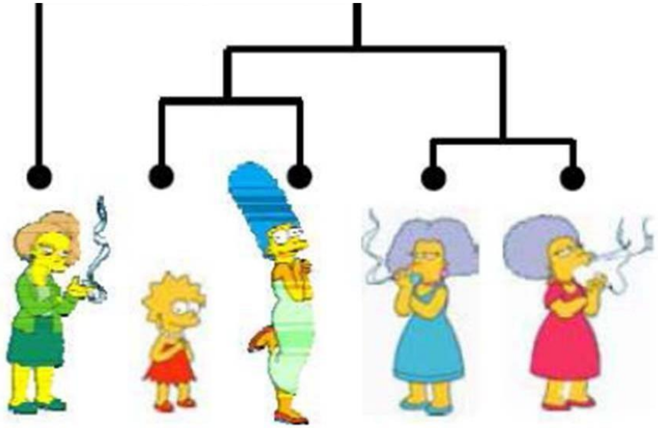
  - Clustering results are crucially dependent on the measure of similarity (or distance) between "points" to be clustered

Clustering algorithms

- Partition algorithms (Flat)
  - K-means
  - Mixture of Gaussian
  - Spectral Clustering



- Hierarchical algorithms
  - Bottom up – agglomerative
  - Top down – divisive



https://www.youtube.com/watch?v=7enWesSofhg
https://www.youtube.com/watch?v=zxQF8Rmpk1M

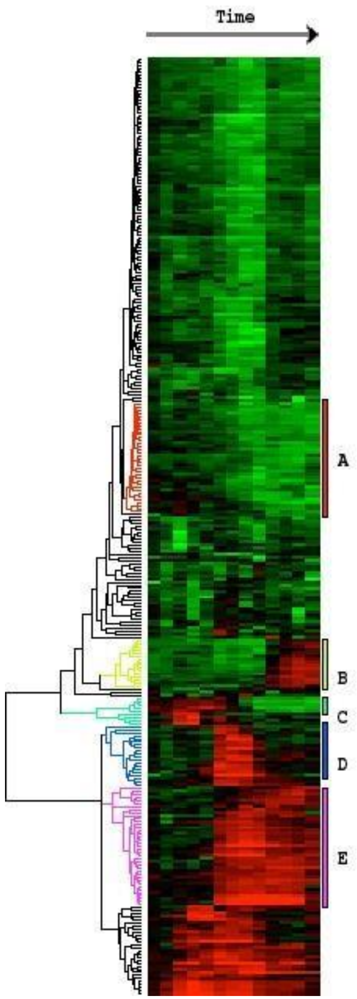
Clustering examples

Image segmentation

Goal: Break up the image into meaningful or perceptually similar regions

Clustering examples

Clustering gene expression data



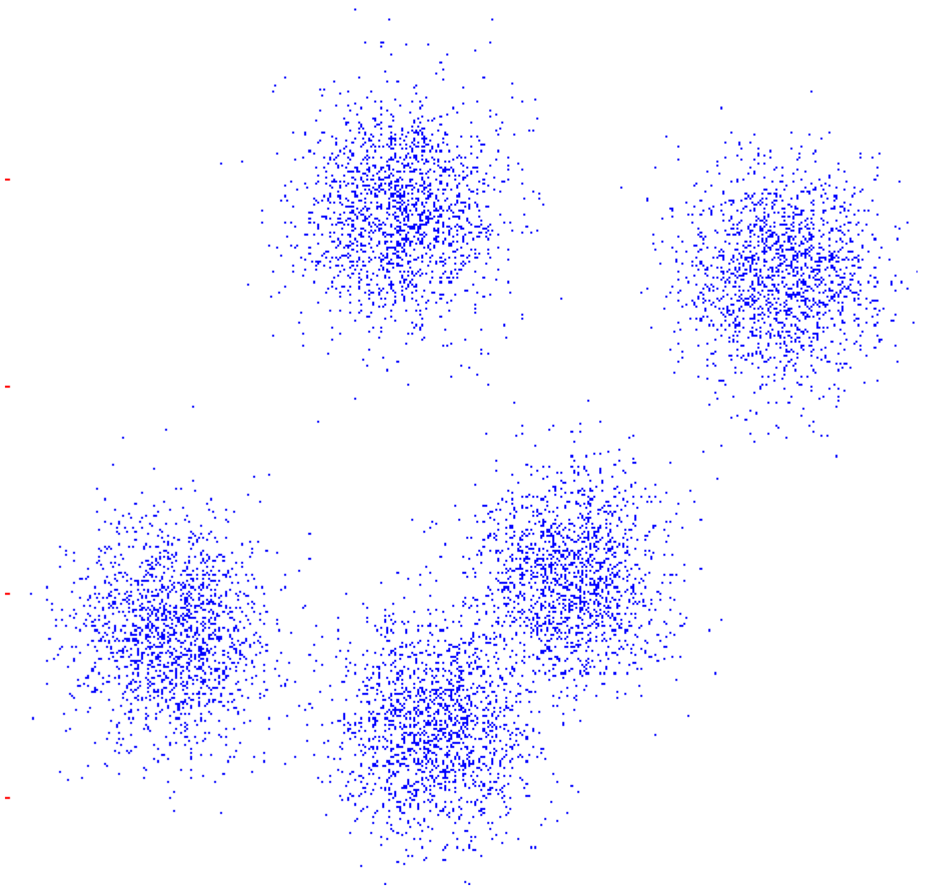
K-Means

- An iterative clustering algorithm

  - Initialize: Pick *K* random points as cluster centers

  - Alternate:
    1. Assign data points to closest cluster center
    2. Change the cluster center to the average of its assigned points

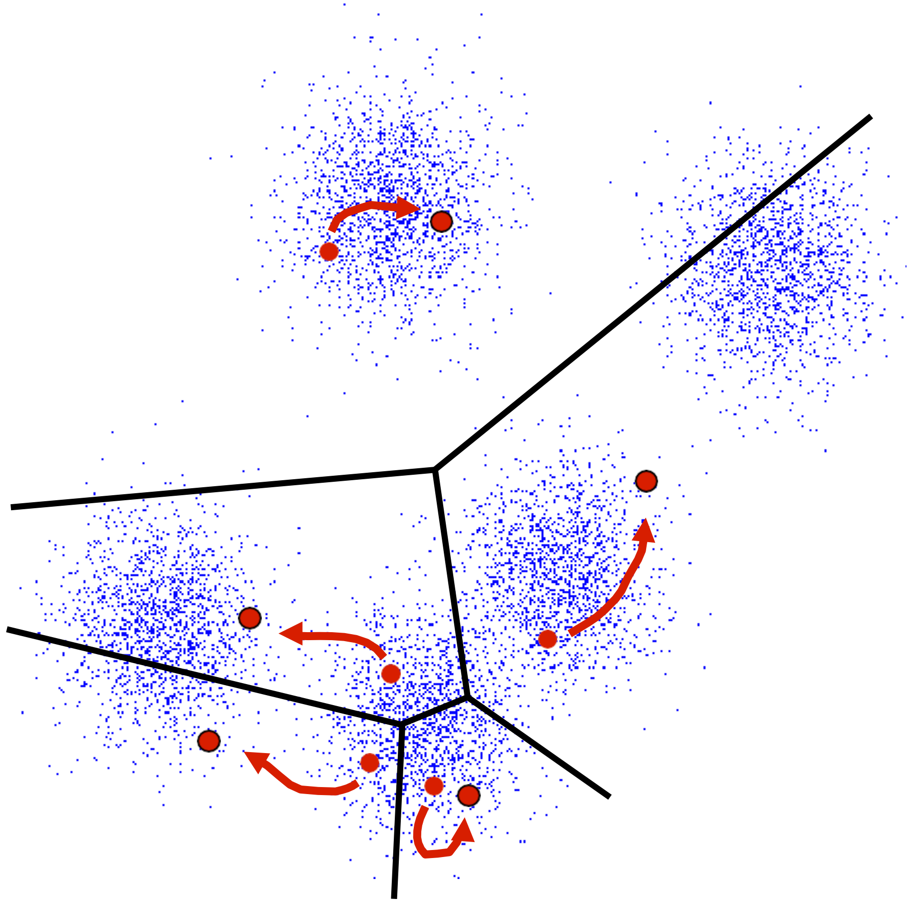
  - Stop when no points' assignments change



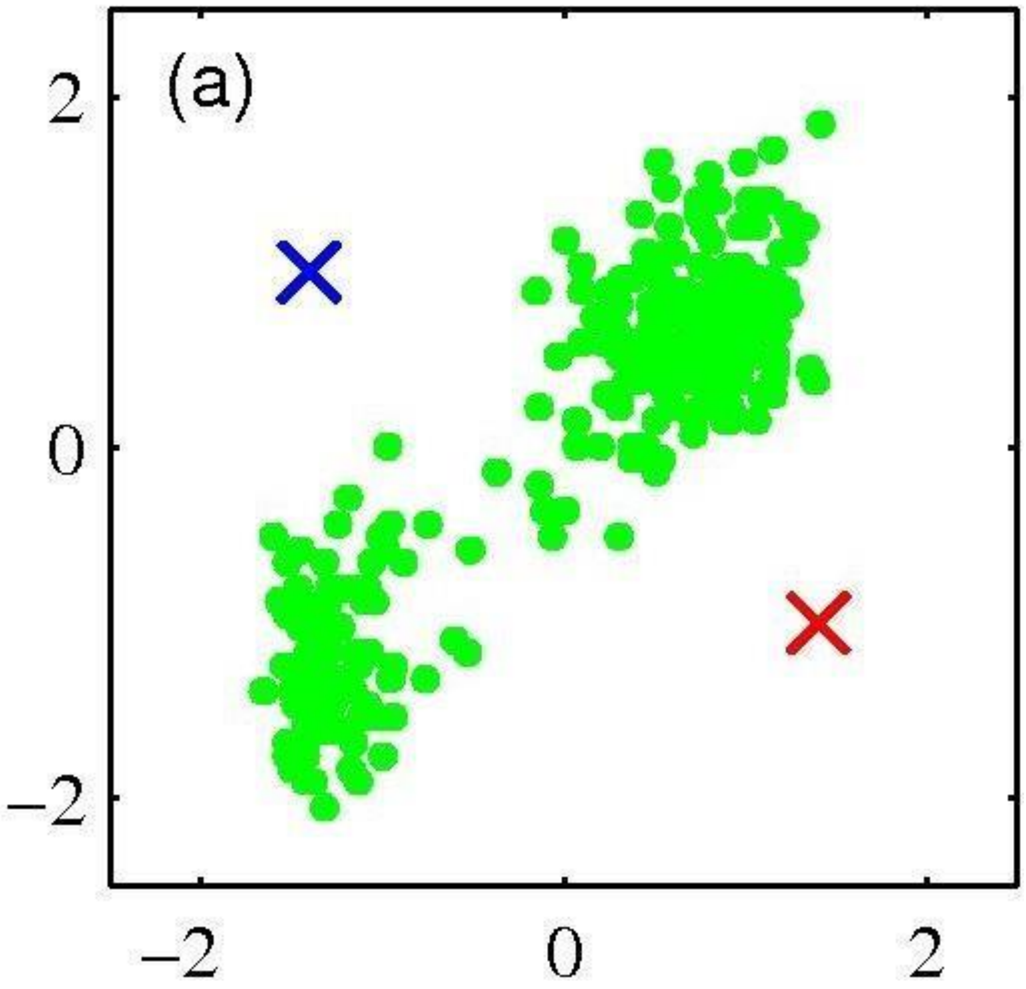
https://www.youtube.com/watch?v=oLp8OQKR_N0

K-Means

- An iterative clustering algorithm
  - Initialize: Pick *K* random points as cluster centers
  - Alternate:
    1. Assign data points to closest cluster center
    2. Change the cluster center to the average of its assigned points
  - Stop when no points' assignments change



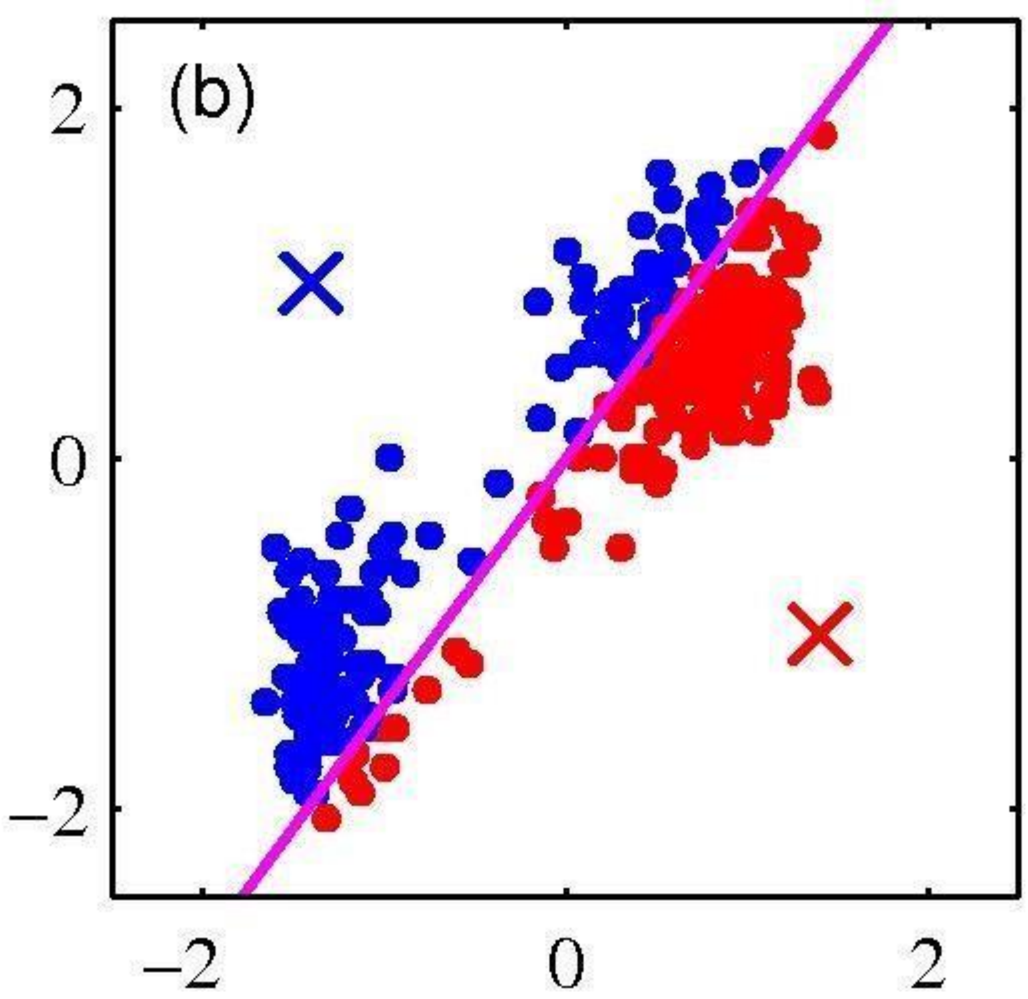
K-means clustering: Example



- Pick K random points as cluster centers (means)

Shown here for K=2

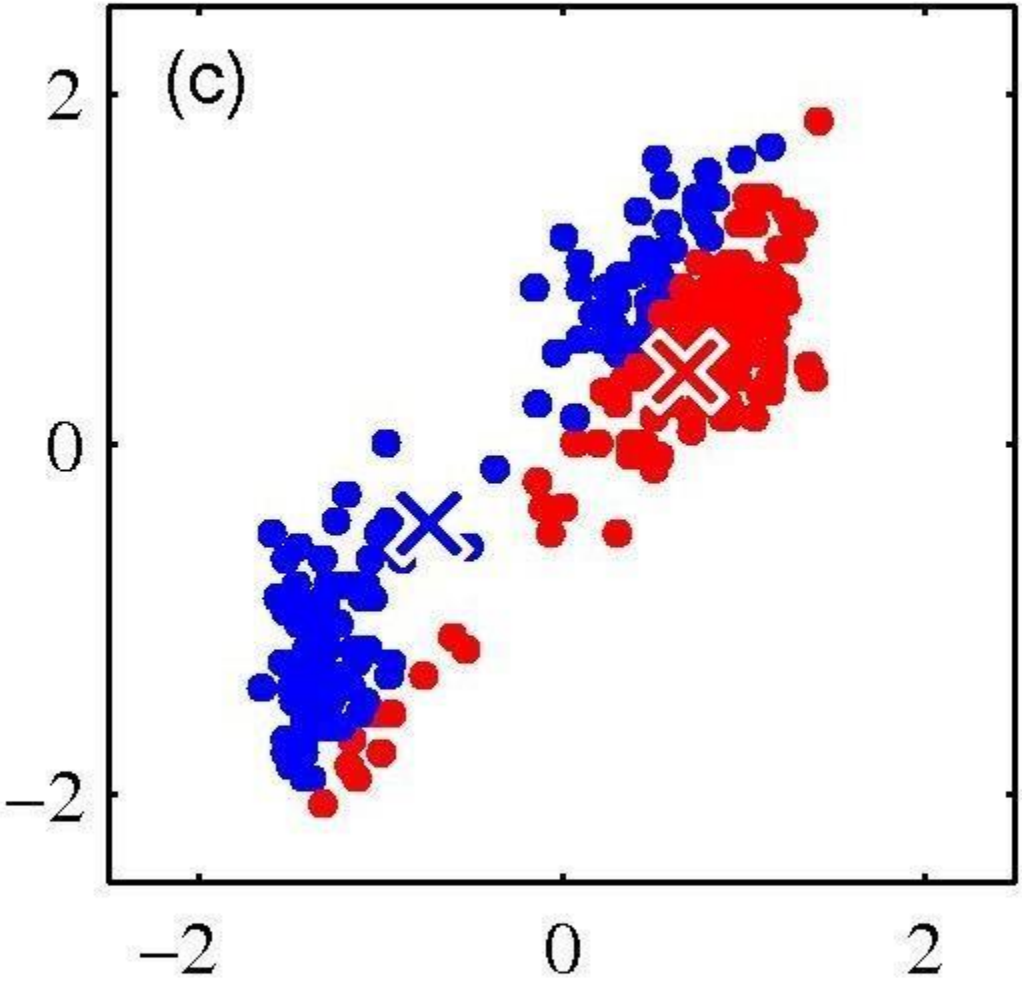
K-means clustering: Example



Iterative Step 1

- Assign data points to closest cluster center

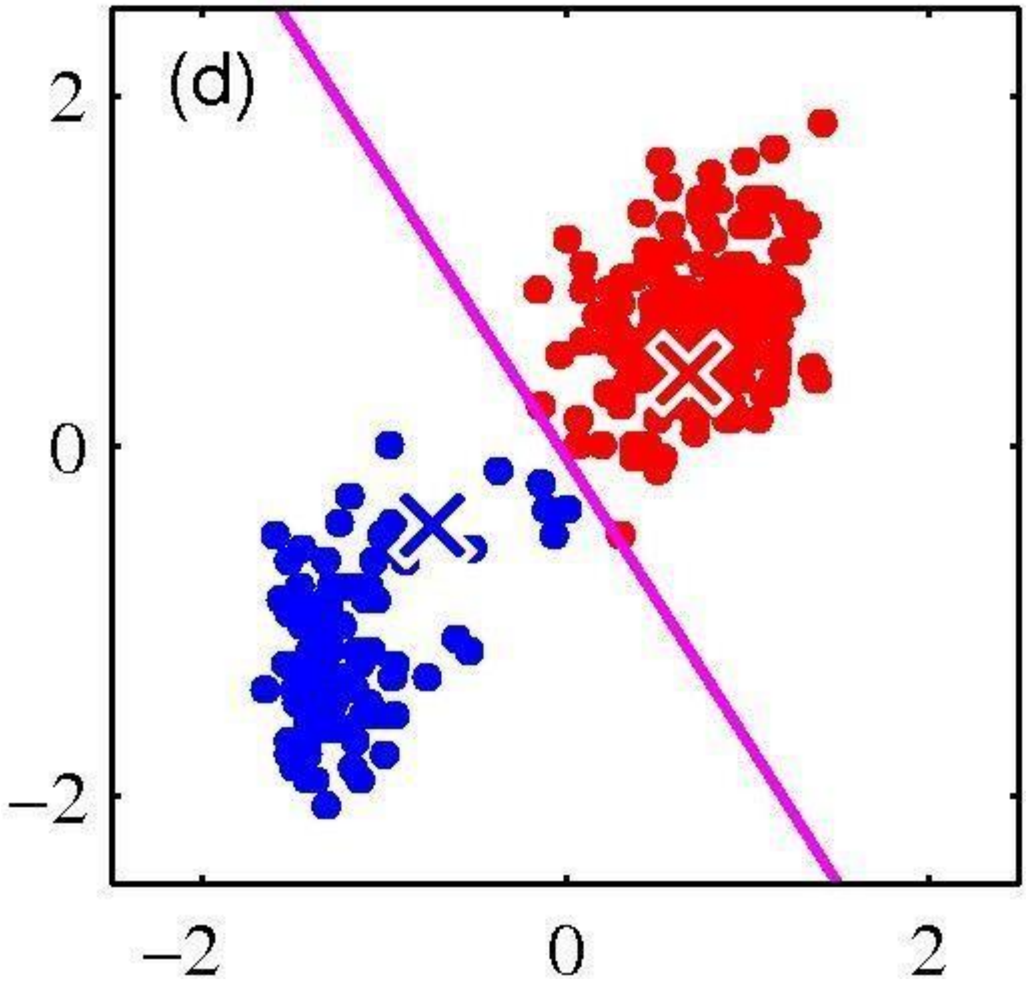
K-means clustering: Example



Iterative Step 2

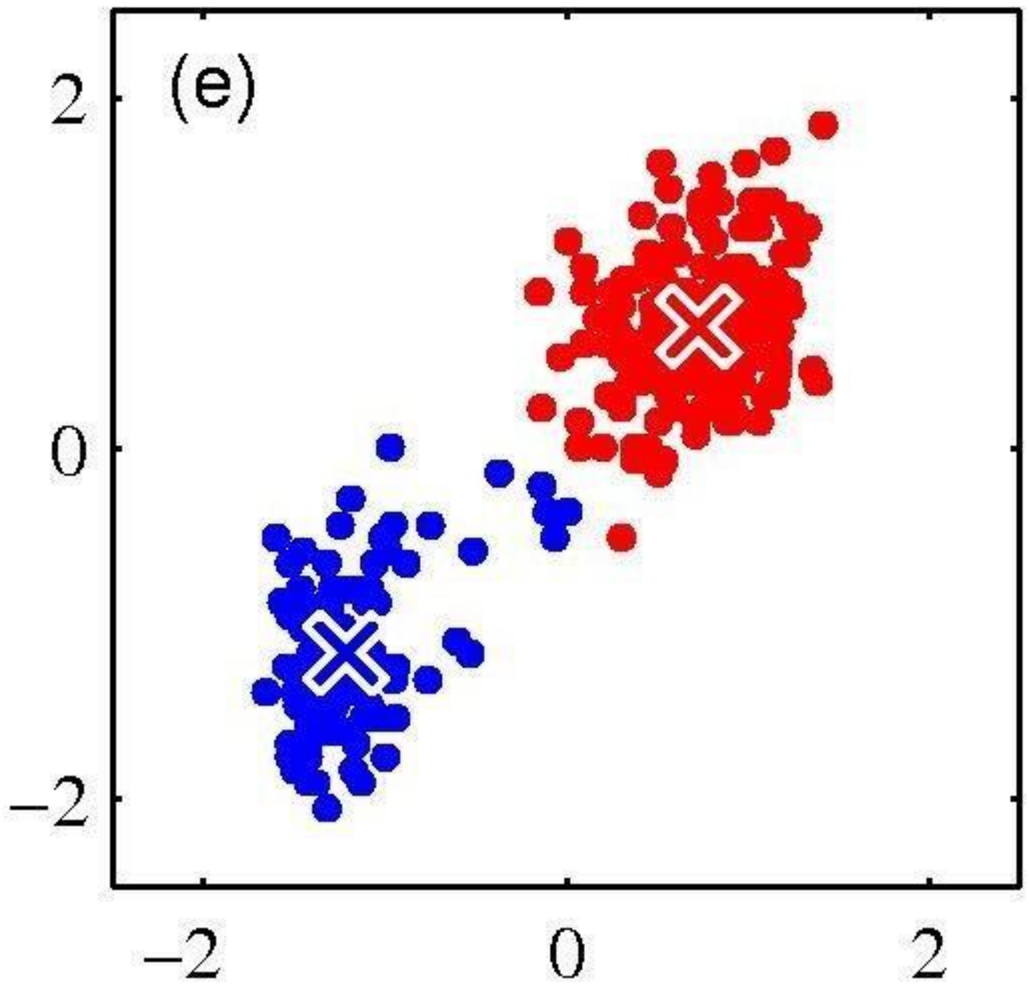
- Change the cluster center to the average of the assigned points

K-means clustering: Example



- Repeat until convergence

K-means clustering: Example



Properties of K-means Algorithm

- Guaranteed to converge in a finite number of iterations

- Running time per iteration:
    1. Assign data points to closest cluster center

       O(KN) time

    2. Change the cluster center to the average of its assigned points

       O(N)

K-Means Clustering – Solved Example

- Suppose that the data mining task is to cluster points into three clusters,
- where the points are
- $A1(2, 10)$, $A2(2, 5)$, $A3(8, 4)$, $B1(5, 8)$, $B2(7, 5)$, $B3(6, 4)$, $C1(1, 2)$, $C2(4, 9)$.
- The distance function is Euclidean distance.
- Suppose initially we assign $A1$, $B1$, and $C1$ as the center of each cluster, respectively.

https://www.youtube.com/watch?v=KzJORp8bgqs

K-Means Clustering – Solved Example

Assign each data point to the nearest cluster centroid

Iteration 1

Initial Centroids:

A1: (2, 10)

B1: (5, 8)

C1: (1, 2)

Data Points			Distance to			Cluster	New Cluster
			2 10	5 8	1 2		
A1	2	10	0.00	3.61	8.06	1	
A2	2	5	5.00	4.24	3.16	3	
A3	8	4	8.49	5.00	7.28	2	
B1	5	8	3.61	0.00	7.21	2	
B2	7	5	7.07	3.61	6.71	2	
B3	6	4	7.21	4.12	5.39	2	
C1	1	2	8.06	7.21	0.00	3	
C2	4	9	2.24	1.41	7.62	2	

$$d(p_1, p_2) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

K-Means Clustering – Solved Example

Initial Centroids:
A1: (2, 10)
B1: (5, 8)
C1: (1, 2)

Data Points			Distance to						Cluster	New Cluster
			2	10	5	8	1	2		
A1	2	10	0.00		3.61		8.06		1	
A2	2	5	5.00		4.24		3.16		3	
A3	8	4	8.49		5.00		7.28		2	
B1	5	8	3.61		0.00		7.21		2	
B2	7	5	7.07		3.61		6.71		2	
B3	6	4	7.21		4.12		5.39		2	
C1	1	2	8.06		7.21		0.00		3	
C2	4	9	2.24		1.41		7.62		2	

New Centroids:
A1: (2, 10)
B1: (6, 6)
C1: (1.5, 3.5)

- Update each cluster centroid μ_j to be the mean of all data points assigned to cluster j:

$$\mu_j = \frac{1}{|S_j|} \sum_{i \in S_j} x_i$$

where S_j is the set of data points assigned to cluster j.

K-Means Clustering – Solved Example

Iteration 2

Current Centroids:

A1: (2, 10)

B1: (6, 6)

C1: (1.5, 3.5)

Assign each data point to the nearest cluster centroid

Data Points			Distance to						Cluster	New Cluster
			2	10	6	6	1.5	1.5		
A1	2	10	0.00		5.66		6.52		1	1
A2	2	5	5.00		4.12		1.58		3	3
A3	8	4	8.49		2.83		6.52		2	2
B1	5	8	3.61		2.24		5.70		2	2
B2	7	5	7.07		1.41		5.70		2	2
B3	6	4	7.21		2.00		4.53		2	2
C1	1	2	8.06		6.40		1.58		3	3
C2	4	9	2.24		3.61		6.04		2	1

$$d(p_1, p_2) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

K-Means Clustering – Solved Example

Current Centroids:
A1: (2, 10)
B1: (6, 6)
C1: (1.5, 3.5)

Data Points			Distance to						Cluster	New Cluster
			2	10	6	6	1.5	1.5		
A1	2	10	0.00		5.66		6.52		1	1
A2	2	5	5.00		4.12		1.58		3	3
A3	8	4	8.49		2.83		6.52		2	2
B1	5	8	3.61		2.24		5.70		2	2
B2	7	5	7.07		1.41		5.70		2	2
B3	6	4	7.21		2.00		4.53		2	2
C1	1	2	8.06		6.40		1.58		3	3
C2	4	9	2.24		3.61		6.04		2	1

New Centroids:
A1: (3, 9.5)
B1: (6.5, 5.25)
C1: (1.5, 3.5)

- Update each cluster centroid μ_j to be the mean of all data points assigned to cluster j:

$$\mu_j = \frac{1}{|S_j|} \sum_{i \in S_j} x_i$$

where S_j is the set of data points assigned to cluster j.

K-Means Clustering – Solved Example

Assign each data point to the nearest cluster centroid

Iteration 3

Current Centroids:

A1: (3, 9.5)

B1: (6.5, 5.25)

C1: (1.5, 3.5)

Data Points			Distance to			Cluster	New Cluster
			3, 9.5	6.5, 5.25	1.5, 3.5		
A1	2	10	1.12	6.54	6.52	1	1
A2	2	5	4.61	4.51	1.58	3	3
A3	8	4	7.43	1.95	6.52	2	2
B1	5	8	2.50	3.13	5.70	2	1
B2	7	5	6.02	0.56	5.70	2	2
B3	6	4	6.26	1.35	4.53	2	2
C1	1	2	7.76	6.39	1.58	3	3
C2	4	9	1.12	4.51	6.04	1	1

$$d(p_1, p_2) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

K-Means Clustering – Solved Example

Current Centroids:
A1: (3, 9.5)
B1: (6.5, 5.25)
C1: (1.5, 3.5)

New Centroids:
A1: (3.67, 9)
B1: (7, 4.33)
C1: (1.5, 3.5)

Data Points			Distance to			Cluster	New Cluster
			3, 9.5	6.5, 5.25	1.5, 3.5		
A1	2	10	1.12	6.54	6.52	1	1
A2	2	5	4.61	4.51	1.58	3	3
A3	8	4	7.43	1.95	6.52	2	2
B1	5	8	2.50	3.13	5.70	2	1
B2	7	5	6.02	0.56	5.70	2	2
B3	6	4	6.26	1.35	4.53	2	2
C1	1	2	7.76	6.39	1.58	3	3
C2	4	9	1.12	4.51	6.04	1	1

- Update each cluster centroid μ_j to be the mean of all data points assigned to cluster j:

$$\mu_j = \frac{1}{|S_j|} \sum_{i \in S_j} x_i$$

where S_j is the set of data points assigned to cluster j.

K-Means Clustering – Solved Example

Assign each data point to the nearest cluster centroid

Iteration 4

Current Centroids:
A1: (3.67, 9)
B1: (7, 4.33)
C1: (1.5, 3.5)

Data Points			Distance to			Cluster	New Cluster
			3.67, 9	7, 4.33	1.5, 3.5		
A1	2	10	1.94	7.56	6.52	1	1
A2	2	5	4.33	5.04	1.58	3	3
A3	8	4	6.62	1.05	6.52	2	2
B1	5	8	1.67	4.18	5.70	1	1
B2	7	5	5.21	0.67	5.70	2	2
B3	6	4	5.52	1.05	4.53	2	2
C1	1	2	7.49	6.44	1.58	3	3
C2	4	9	0.33	5.55	6.04	1	1

After four updates of centroid the final clusters are:
Cluster 1: A1, B1, C2
Cluster 2: A3, B2, B3
Cluster 3: A2, C1

$$d(p_1, p_2) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Example 2: K-Means

Solve using Kmean Clustering

$K = \{2, 3, 4, 10, 11, 12, 20, 25, 30\}$

Consider 2 clusters ($K=2$)

$\Rightarrow$ $m_1: 4$ $\quad m_2: 12$

Step1: $K_1 = \{2, 3, 4\}$ $\quad K_2 = \{10, 11, 12, 20, 25, 30\}$

$m_1: \frac{2+3+4}{3} = \frac{9}{3} = 3$ $\quad m_2 = \frac{108}{6} = 18$

Step2: $K_1 = \{2, 3, 4, 10\}$ $\quad K_2 = \{11, 12, 20, 25, 30\}$

$m_1: 4.75$ $\quad m_2 = 19.6$

Step3: $K_1 = \{2, 3, 4, 10, 11, 12\}$

$K_2 = \{20, 25, 30\}$

$m_1: 7$ $\quad m_2: 25$

Step4: $K_1 = \{2, 3, 4, 10, 11, 12\}$

$K_2 = \{20, 25, 30\}$

$m_1 = 7$ $\quad m_2 = 25$

We are getting the same means,
so the new clusters are:
Cluster 1: K1 = {2, 3, 4, 10, 11, 12}
Cluster 2: K2 = {20, 25, 30}

https://www.youtube.com/watch?v=TsjXvANsN0c
https://www.youtube.com/watch?v=YWgcKSa_2ag

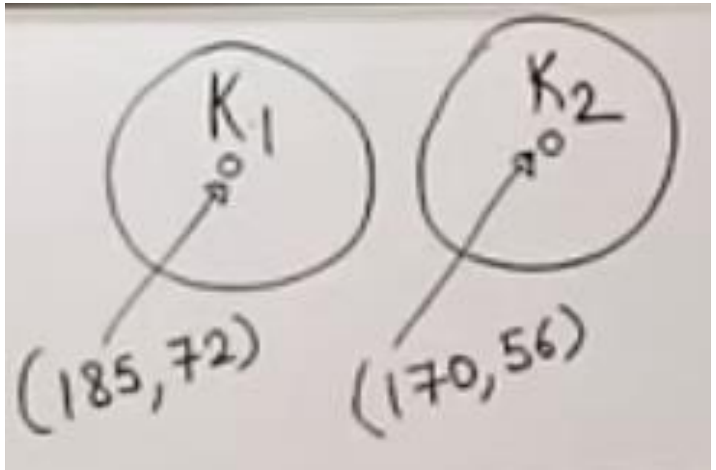
Example 3: K-Means

	Height	weight
1	185	72
2	170	56
3	168	60
4	179	68
5	182	72
6	188	77
7	180	71
8	180	70
9	183	84
10	180	88
11	180	67
12	177	76

Euclidean Distance

$$\sqrt{(X_o - X_c)^2 + (Y_o - Y_c)^2}$$

O = Observed
C = Centeroid



https://www.youtube.com/watch?v=CLKW6uWJtTc

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ED for 3 $\rightarrow K_1 \rightarrow \sqrt{(168-185)^2+(60-72)^2}$

20.80

$\rightarrow K_2 \rightarrow \sqrt{(168-170)^2+(60-56)^2}$

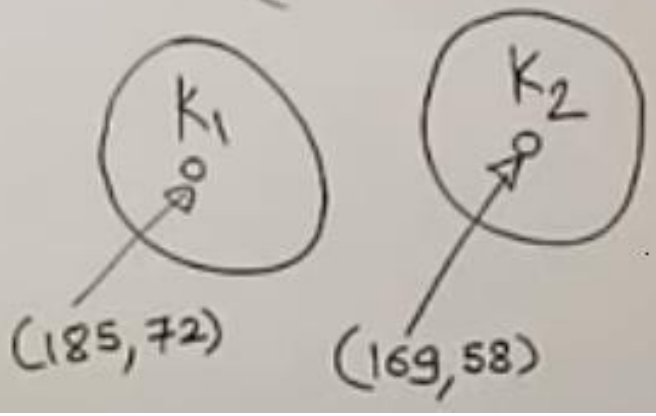
$= 4.48$

Example 3: K-Means

	Height	weight
1	185	72
2	170	56
3	168	60
4	179	68
5	182	72
6	188	77
7	180	71
8	180	70
9	183	84
10	180	88
11	180	67
12	177	76

New Centroid Calculation :-

for $K_2 = \left(\frac{170+168}{2}, \frac{60+56}{2}\right) = (169, 58)$



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	Height	weight
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6	188	77
7	180	71
8	180	70
9	183	84
10	180	88
11	180	67
12	177	76

E.D for → $k_1 = \sqrt{(179-185)^2+(68-72)^2}$

4 $= (6.32)$

→ $k_2 = \sqrt{(179-169)^2+(68-58)^2}$

$= 14.14$

Find distance with each point and then assign them to the nearest cluster based on the distance value, update the centroid value by calculating the mean when a new point is added into it.

Following are the final points assigned to two clusters.

$K_1 \rightarrow \{1,4,5,6,7,8,9,10,11,12\}$

$K_2 \rightarrow \{2,3\}$